Tr. Julius.W.M 0787093081

Candidate's Name: SAMPLE SCORE ……..

Signature: …….. | **STREAM:** | | | | | | | |

525/1
CHEMISTRY
Paper 1
Nov./Dec.
2025
$2\frac{1}{2}$ hours

END OF YEAR ASSESSMENT 2025

Uganda Advanced Certificate of Education

S.5 CHEMISTRY

Paper 1

(*Set I*)

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

This paper has **two** *sections* **A** *and* **B.**

Section **A** *has two compulsory items while* **B** *has two parts* **Part I** *and* **Part II**

Each of part I and part II has two items, Answer only **one** *from each.*

Answers to Section **A** ***must*** *be written in the spaces provided and Section* **B** ***must*** *be written in the answer booklet(s) provided*

Answer four in all.

Where necessary use,

Molar gas volume at s.t.p = 22.4dm^3

FOR EXAMINER'S USE ONLY				
ITEM	**CODE**		**SCORE**	
1	F_1		04	
	C_1		04	
	C_2		06	
	C_3		03	
	D_1		03	
2	E_c		06	
	C_4		02	
	C_5		04	
	E_1		05	
	D_2		03	
3/4	G_r	V_p	05	08
	T_r	R	08	09
	E_v	D_3	04	03
	P_r		03	
5/6	F_2		06	
	Mc		02	
	S		11	
	D_3		03	
TOTAL			80	

Key

C_1, C_2, C_3, C_4, C_5: *Calculations*
D_1, D_2, D_3: *Danger*
E_c: *Energy Cycle*
E_V: *Evaluation*
G_r: *Graphing*
T_r: *Trend*
P_r: *Pridiction*
F: *Formula*
Mc: *Mechanism*
S: *Synthesis*
V_p: *VSEPR theory*
R: *Reason*

SECTION A

Answer **all** *questions in this section*

ITEM 1

During a rapid response investigation near Kitezi landfill, the Water Quality Surveillance Unit discovered heavy metal contamination in a borehole supplying a large number of households.

A fragment of metallic lead from a nearby illegal battery recycling site was analysed using a mass spectrometer. The detector registered four isotopic signals with currents of *0.16 mA (mass 204), 2.72 mA (mass 206), 2.50 mA (mass 207)* and *5.92 mA (mass 208)*. The investigation team was briefed that these readings can help determine an accurate relative atomic mass (RAM) of lead.

Further laboratory tests on the contaminated water revealed that the dissolved poisonous salt responsible for the lead toxicity contained *62.6%* lead, *8.45%* nitrogen, and the rest oxygen molar mass of the salt was reported as *331.22* g/mol.

The concentration of the dissolved salt in the borehole water was measured as 0.90 mol dm^{-3}, and the borehole supplies approximately 1200 litres of water per day. Chemists concluded that this salt can be removed by treating the water with sodium sulphate solution, which reacts to produce insoluble lead(II) sulphate.

In extreme contamination cases, *0.45 moles* of nitrogen dioxide gas was released during side reactions at 28°C and 98 kPa. To protect public health, the Ministry has assigned your team to analyse the data and provide a complete stoichiometric and methodological evaluation of the situation.

Task:

As a learner of Chemistry, help the company to know;

(a) The relative atomic mass of lead. **(RAM)**.

$\boldsymbol{Total\ current = 0.16 + 2.72 + 2.50 + 5.92 = 11.3mA}$ ✓CA

$\boldsymbol{Percentage\ abundance\ of\ lead\ -\ 204 = \left(\frac{0.16}{11.3} \times 100\right) = 1.42}$

$\boldsymbol{Percentage\ abundance\ of\ lead\ -\ 206 = \left(\frac{2.72}{11.3} \times 100\right) = 24.07}$

$\boldsymbol{Percentage\ abundance\ of\ lead\ -\ 207 = \left(\frac{5.92}{11.3} \times 100\right) = 22.12}$ ✓CA

$\boldsymbol{Percentage\ abundance\ of\ lead\ -\ 208 = \left(\frac{2.72}{11.3} \times 100\right) = 52.40}$

$\boldsymbol{R.A.M = \sum Relative\ isotopic\ mass \times proportion\ of\ isotope}$

$\boldsymbol{R.A.M = \left(204 \times \frac{1.42}{100}\right) + \left(206 \times \frac{24.07}{100}\right) + \left(207 \times \frac{22.12}{100}\right) + \left(208 \times \frac{52.4}{100}\right)}$ ✓CA

$\mathbf{R.A.M = 207.26}$ ✓CA

$4CA = C_1 = 04\ scores$

(b) To determine the molecular formula of the poisonous salt in water.

$Lead = 62.6\%, Nitrogen = 8.45\%, Oxygen = 100 - (62.6 + 8.45) = 28.95\%$ CA

Element	Pb	N	O
%composition	62.6	8.45	28.95
Moles	$\frac{62.6}{207.26} = 0.302$	$\frac{8.45}{14} = 0.604$	$\frac{28.95}{16} = 1.809$ CA
Simplest ratio	$\frac{0.302}{0.302} = 1$	$\frac{0.604}{0.302} = 2$	$\frac{1.809}{0.302} = 6$

$Empirical\ formula\ of\ salt\ is\ PbN_2O_6$ F

$(PbN_2O_6)n = 303.3$

$(207.26 \times 1n) + (14 \times 2n) + (16 \times 6n) = 331.22 \quad \therefore n = 1$

$Molecular\ formula\ of\ the\ salt\ is\ Pb(NO_3)_2$ F Name: $Lead(ii)nitrate$ NA

$2CA + 2F + NA = F_1 = 04\ Scores$

(c) To calculate the mass of lead(II)sulphate formed if all the dissolved salt was removed

$0.9moll^{-1}means;\ 1litre\ of\ borehole\ water\ contain\ 0.9moles\ of\ Pb(NO_3)_2$ CA

$1200litres\ of\ borehole\ water\ contain\ \left(\frac{0.9\times1200}{1}\right)$ CA

$= 1080moles\ of\ Pb(NO_3)_2$

$From\ equation: Pb(NO_3)_2(aq) + Na_2SO_4(aq) \longrightarrow PbSO_4(s) + 2NaNO_3(aq)$ CA

$1mole\ of\ Pb(NO_3)_2\ produces\ 1mole\ of\ PbSO_4$

$1080moles\ of\ Pb(NO_3)_2\ produce\ \left(\frac{1}{1}\times1080\right)moles\ of\ PbSO_4. = 1080moles$ CA

$Molar\ mas\ of\ PbSO_4\ is\ (207 \times 1) + (32 \times 1) + (16 \times 4) = 303g$ CA

$1mole\ of\ PbSO_4\ weigh\ 303g$

$\therefore 1080moles\ of\ PbSO_4\ weigh\ \left(\frac{303\times1080}{1}\right) = 327240g$ CA $or\ 3.27 \times 10^5 kg\ of\ PbSO_4$ CA

$7CA = C_2 = 06\ scores$

(d) Know the volume of gas produced using the gas law.

$From\ gas\ law: PV = nRT; P = 98 \times 10^3 pa, T = (28 + 273) = 301k\ and\ R = 8.314$ CA

Note: *We use*;

1. $R = 0.0821\ \text{L} \cdot \text{atm}/(\text{mol} \cdot \text{K}), when, P\ is\ in\ atm\ and\ V\ is\ in\ litre/dm^3$
2. $R = 8.314\ J/(molK), when, P\ is\ in\ Pa\ and\ Volume\ is\ in\ m^3$

$\therefore V = \frac{nRT}{P} = \left(\frac{0.45\times8.314\times301}{98\times10^3}\right)$ CA

$= 0.0115m^3$ CA

$3CA = C_3 = 06\ scores$

(e) Environmental impacts of the gaseous or aqueous system and their mitigations.

*Nitrogen dioxide contributes to **global warming** when it accumulates in the atmosphere, since it traps heat and creating the greenhouse effect, leading to rising temperatures hence **drought**. This can be mitigated by using catalytic converters to reduce it into less harmful substances such as nitric acid.*

*Nitrogen dioxide causes **respiratory problems** and **poor air quality.** When inhaled, it irritates the lungs and reduces oxygen uptake, increasing cases of **asthma** and **bronchitis.** Using personal protective gears like gas masks during scientific investigations.*

$di + de + dm = D_1 = 03scores$

Item 2

A recent report from Tororo Industrial Chemicals Ltd, one of Uganda's emerging producers of industrial reagents and laboratory kits, revealed that the company is developing a dual-purpose field chemical pack for technicians working in rural agricultural and health sectors. The pack is intended to include one safe *warming* substance for sterilising small instruments and generating heat in remote operations, and one *cooling* substance for preserving biological samples and stabilising temperature sensitive reactions.

The company's research department is currently *comparing* ethanol, a locally available biofuel derived from sugarcane waste, with *sodium chloride*, obtained from the salt works around Lake Katwe.

The thermochemical data obtained from a certain article about ethanol indicate that the standard heats of formation of ethanol, carbon dioxide and water are (−227.0) , (−393.5), (−285.5) kJ mol^{-1} respectively.

For sodium chloride, the enthalpy of sublimation of sodium, bond dissociation energy of chlorine, first ionisation energy of sodium, first electron affinity of chlorine, enthalpy of formation of sodium chloride, hydration enthalpy of $Na^+(aq)$ and hydration enthalpy of $Cl^-(aq)$,(+109), (+242), (+494), (−364), (−411), (−406), and (−363) kJ mol^{-1} respectively.

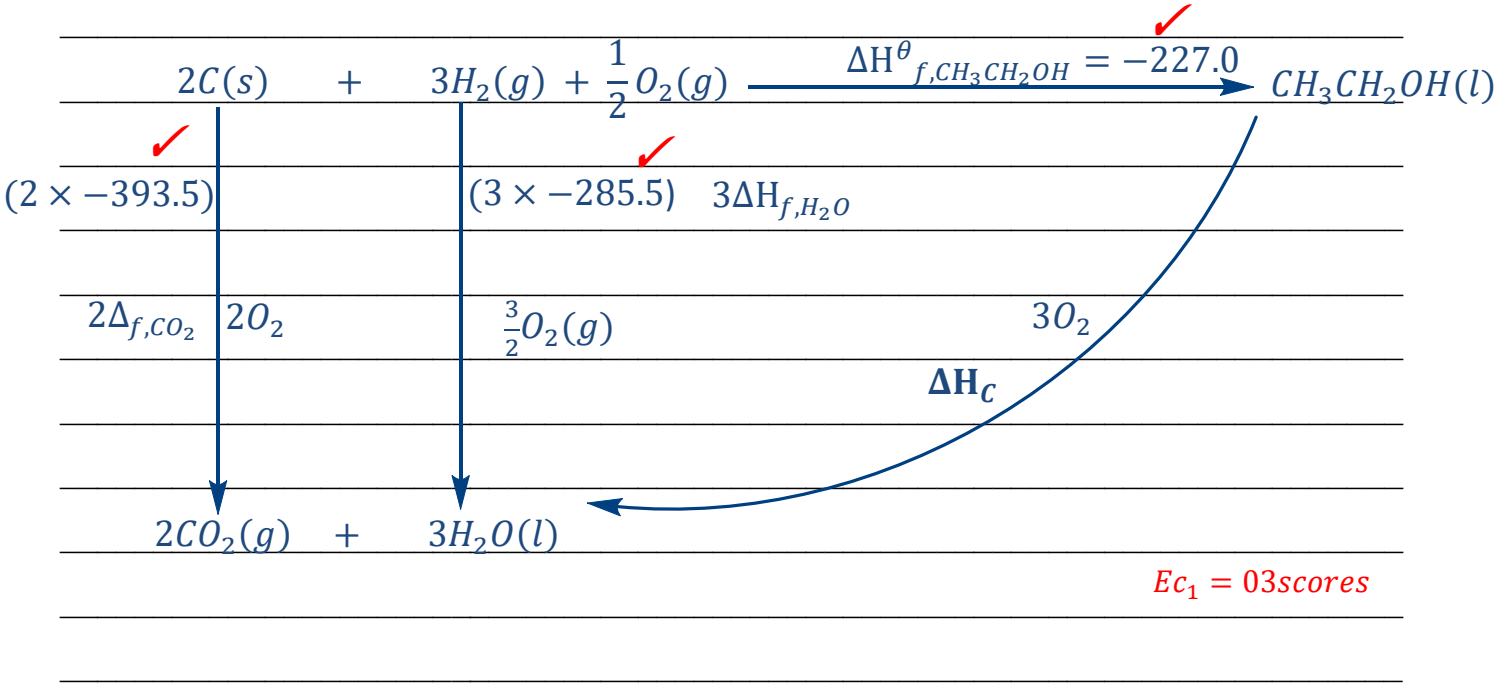
Preliminary tests by the plant's laboratory team indicated that combustion of ethanol and, dissolution of sodium chloride in water produces minimal temperature change. To ensure the most energy-efficient and environmentally safe field pack, the company manager has invited you.

Task

As a student of chemistry, help the researcher;

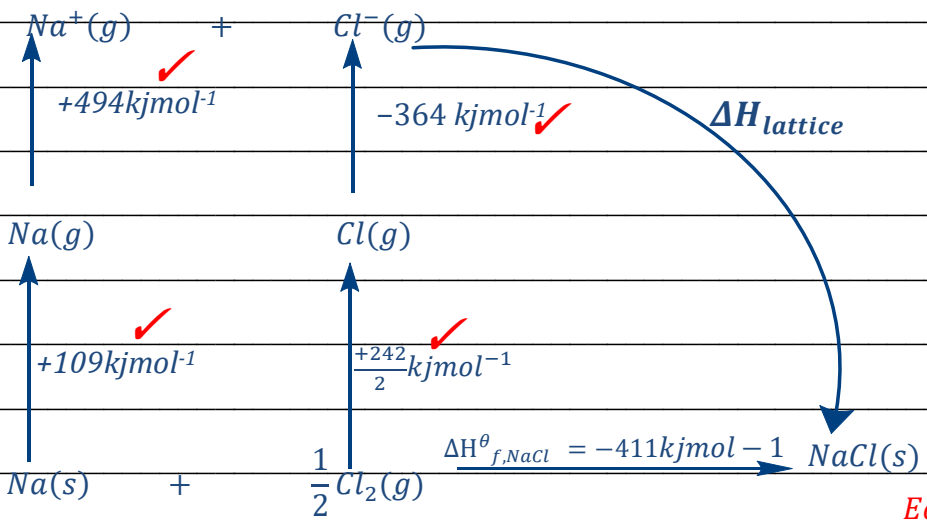
(a) Analyse the data to construct an energy cycle for the changes in the:

(i) Combustion of ethanol.



$Ec_1 = 03scores$

(ii) Formation of sodium chloride.



$Ec_2 = 03scores$

$Ec_1 + Ec_2 = E_c = 06\ scores$

(b) Use the energy cycles constructed to calculate;

(i). Enthalpy of combustion of ethanol.

By Hess' law;

$$2\Delta_{f,CO_2} + 3\Delta H_{f,H_2O} = \Delta H^{\theta}{}_{f,CH_3CH_2OH} + \boldsymbol{\Delta H_C}$$

$$\boldsymbol{\Delta H_C} = 2\Delta_{f,CO_2} + 33\Delta H_{f,H_2O} - \Delta H^{\theta}{}_{f,CH_3CH_2OH}$$

$$= (2 \times -393.5) + (3 \times -285.5) - (-227.0)$$ CA

$$\boldsymbol{\Delta H_C = -1416.5 kjmol^{-1}}$$ CA

$2CA = C_4 = 02\ Scores$

(ii). Lattice of sodium chloride and its enthalpy of solution.

By Hess' law;

$$\Delta H^{\theta}{}_{f,NaCl} = \Delta H^{\theta}{}_{sublimatio} + \Delta H^{\theta}{}_{1st\ IE} + \frac{1}{2}\Delta H^{\theta}{}_{BDE} + \Delta H^{\theta}{}_{HA} + \boldsymbol{\Delta H_{lattice}}$$

$$\boldsymbol{-411 = 109 + 494 + \frac{1}{2} \times 242 + -364 + \Delta H_{lattice}}$$ CA

$$\boldsymbol{\Delta H_{lattice} = -771 kjmol^{-1}}$$ CA

$$\Delta H_{solution\ Nacl} = +\Delta H_{lattice\ Nacl} + +\Delta H_{hydration\ Nacl}$$

$$\Delta H_{solution\ Nacl} = +771 + \left(\Delta H_{hydration\ Na^+} + \Delta H_{hydration\ Cl^-}\right) = +771 + (-406 + -363)$$

$$\Delta H_{solution\ NaCl} = +2kjmol^{-1}$$ CA

CA

$4CA = C_5 = 04\ Scores$

(c) (i) Evaluate industrial usefulness of the two systems to advise the manager choose the best substance for warming, and cooling with reason.

Er

Ev

Ethanol is more suitable for warming functions; due its ***high exothermic*** *enthalpy of combustion of* $-1416.5kjmol^{-1}$, *this energy is supplied for heating up other substances.*

Whereas;

Ev

Er

Sodium chloride is more suitable for cooling functions; due to its ***endothermic*** *enthalpy of solution* $+2kjmol^{-1}$, *during solution of sodium chloride, heat is absorbed from other substances leaving them cooler.*

(ii). State any other ideal industrial use of ethanol.

U

Used as a ***solvent*** *for extracting perfumes, drugs, plant essences, and paints since it acts as both polar and non-polar solvent.*

U

70% ethanol is used in ***hand sanitizers*** *and medical wipes since it kills bacteria and viruses on skin and surfaces.*

$2Ev + 2Er + 1U = E_{v1} = 05\ Scores$

(d) Suggest the possible impacts and mitigations of the substances to the environment.

di

de

When ethanol burns, it produces ***carbon dioxide (CO_2)***, *a greenhouse gas that contributes to* ***global warming*** *if released in large quantities resulting to increased drought. mitigated by* ***adopting clean energy technologies*** *such as solar energy systems to reduce emissions.*

dm

Excessive discharge of NaCl brine from industries increases ***salinity*** *in soil and water thus soil and water polluttion. Mitigated by properly disposing brine wastes.*

$1di + 1de + 1dm = D_2 = 03\ Scores$

SECTION B
Part I
Attempt **One** *item in this section*

Item 3
A chemical manufacturing company in Uganda, Kyoga Chemicals Ltd, produces industrial chemicals and water treatment agents. The company has been advised that substances suitable for use in water treatment plants and in construction industries, for stabilizing concrete and producing heat resistant materials must have the strongest ionic characters. The engineering department is analyzing the melting points of chlorides of Group II and Period 3 elements to select materials suitable for operations. The data is shown below:

Group/period	**Group II chlorides**					**Period 3 chlorides**							
Element	$BeCl_2$	$MgCl_2$	$CaCl_2$	$SrCl_2$	$BaCl_2$	NaCl	$MgCl_2$	$AlCl_3$	$SiCl_4$	PCl_3	PCl_5	S_2Cl_2	Cl_2
Atomic No	4	12	20	38	56	11	12	13	14	15	15	16	17
Melting pt/°C	405	714	782	875	962	808	714	192	-68	-92	160	-76	-80

The engineering department seeks a scientific analysis to select materials for different intended purpose. The company is also considering an unknown element X, positioned between aluminium chloride and silicon chloride in the periodic table. *Plotting graphs* of melting point against atomic number for Group II and Period 3 chlorides (separately), *describing and explaining* trends and irregularities in melting points across Period 3 and down Group II chlorides, *evaluating, with reasons*, which chlorides are most suitable for water treatment and stabilizing concrete, casting molds, and producing heat resistant materials and *predicting* the likely melting point range and industrial suitability of the element X chloride.

Task

As a chemistry student, make a write up you will use to help the company.

Item 4
Kilembe Chemical Works Ltd, in Kasese, is investigating how bonding and molecular structure affect the properties and industrial uses of several compounds used in fertilizer formulation, water purification, and manufacturing of plastics and ceramics.

The *compounds* under study include *water*, *phosphorus trichloride*, *carbon tetrachloride*, *tin(II) chloride*, *ammonia*, *tin(IV) chloride*, *nitrate ion*, and *phosphorus pentachloride*.

It was reported that some compounds such as carbon tetrachloride and phosphorus trichloride are liquids at room temperature and insoluble in water, ammonia and water are highly soluble and have strong intermolecular attractions. tin(II) chloride conducts *electricity* in molten form, while tin(IV) chloride does not.

The manager seeks to know *molecular shape*, *bond type*, and *bond angle* for each substance, use bond type and molecular structure in the compounds *account for their differences* in melting point, solubility, and electrical conductivity. why water and ammonia form hydrogen bonds, and how this affects their boiling points and solubility compared to phosphorus trichloride and phosphorus pentachloride. Compare the bonding and structure of tin(II) chloride and tin(IV) chloride explain their difference in electrical conductivity and melting points.to guide the design of safe storage systems and efficient cooling and purification processes. Evaluate the environmental impacts of using nitrate ions in fertilizers

Task

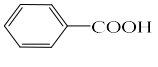
As a chemistry student, make a write up you will use to help the company.

Part II

Attempt **One** *item in this section*

ITEM 5

Nile Chemical Industries Ltd, a chemical manufacturing firm located in Jinja, produces a range of organic-based materials used in plastics, fuels, solvents, and disinfectants. In one of its experiments, 20.0 cm^3 of a gaseous hydrocarbon W was exploded with 150 cm^3 of excess oxygen. After complete combustion, the residual gas measured 110 cm^3, and when concentrated potassium hydroxide solution was added, the volume further decreased to 30 cm^3

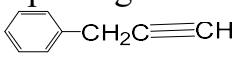
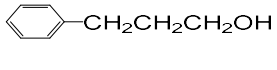
The company noted that ozonolysis of W produced only one compound and that W readily decolourised bromine water. These findings prompted the research team to seek a deeper understanding of W and its possible industrial applications in the synthesis of propan-1-ol, benzoic acid, {  }, 1,2-dibromopentane and 1-chlorobutane.

The company seeks a comprehensive evaluation: to determine the *formulae* of W, draw and name all possible *isomers*, *identify* W with reason(s), and illustrate the mechanism for the reaction between W and bromine water. Additionally, confirm the identity of the gas absorbed by potassium hydroxide solution, *predict* the functional groups of resulting organic compounds, describe *synthetic* pathways to related products, and assess the *environmental* impacts of producing above compounds in industry. You have been contacted for assistance.

Task

As a student of Chemistry, prepare a presentation you will use upon invitation.

ITEM 6

Albertine Oil Fractionation Ltd, an industrial chemical company in the Albertine region, produces fuels, solvents, and organic feedstocks for plastics and chemical synthesis. During a research study, an organic compound P was isolated from a distillation fraction. Complete combustion of a sample of P yielded 8.8 g carbondioxide and 1.8 g water, while 0.100 g of P, vapourised at 273°C and 734 mmHg, occupied 4.46×10^{-2} dm^3. Ozonolysis of P followed by hydrolysis produced a mixed Q of two different compounds, and P was observed to react readily with bromine and concentrated sulphuric acid. The company is exploring industrial applications of these compounds for synthesis of prop-2-yn-1-ylbenzene{  } a component in manufacture of new drugs, benzene{  }solvent, and {  }chemical feedstocks.

The manager seeks to know the formulae of P, all possible structural *isomers*, *identity* of P with reasons, deduce the structure and functional groups of mixture Q, illustrate *mechanisms* for reactions of P with *bromine* and *concentrated sulphuric acid*, , outline *synthetic* pathways from Albertine oil fractions P to synthesis industrial compounds stated, and evaluate the environmental impacts of producing compounds in industry and their mitigations. You have been invited to help the manager.

Task

As a student of Chemistry, prepare a presentation you will use to help the firm.

THE PERIODIC TABLE

1	2											3	4	5	6	7	8
1.0 H 1																1.0 H 1	4.0 He 2
6.9 Li 3	9.0 Be 4											10.8 B 5	12.0 C 6	14.0 N 7	16.0 O 8	19.0 F 9	20.2 Ne 10
23.0 Na 11	24.3 Mg 12											27.0 Al 13	28.1 Si 14	31.0 P 15	32.1 S 16	35.4 Cl 17	40.0 Ar 18
39.1 K 19	40.1 Ca 20	45.0 Sc 21	47.9 Ti 22	50.9 V 23	52.0 Cr 24	54.9 Mn 25	55.8 Fe 26	58.9 Co 27	58.7 Ni 28	63.5 Cu 29	65.7 Zn 30	69.7 Ga 31	72.6 Ge 32	74.9 As 33	79.0 Se 34	79.9 Br 35	83.8 Kr 36
85.5 Rb 37	87.6 Sr 38	88.9 Y 39	91.2 Zr 40	92.9 Nb 41	95.9 Mo 42	98.9 Te 43	101 Ru 44	103 Rh 45	106 Pd 46	108 Ag 47	112 Cd 48	115 In 49	119 Sn 50	122 Sb 51	128 Te 52	127 I 53	131 Xe 54
133 Cs 55	137 Ba 56	139 La 57	178 Hf 72	181 Ta 73	184 W 74	186 Re 75	190 Os 76	192 Ir 77	195 Pt 78	197 Au 79	201 Hg 80	204 Ti 81	207 Pb 82	209 Bi 83	209 Po 84	210 At 85	222 Rn 86
223 Fr 87	226 Ra 88	227 Ac 89															
			139 La 57	140 Ce 58	141 Pr 59	144 Nd 60	147 Pm 61	150 Sm 62	152 Eu 63	157 Gd 64	159 Tb 65	162 Dy 66	165 Ho 67	167 Er 68	169 Tm 68	173 Yb 70	175 Lu 71
			227 Ac 89	232 Th 90	231 Pa 91	238 U 92	237 Np 93	244 Pu 94	243 Am 95	247 Cm 96	247 Bk 97	251 Cf 98	254 Es 99	257 Fm 100	256 Md 101	254 No 102	260 Lw 103

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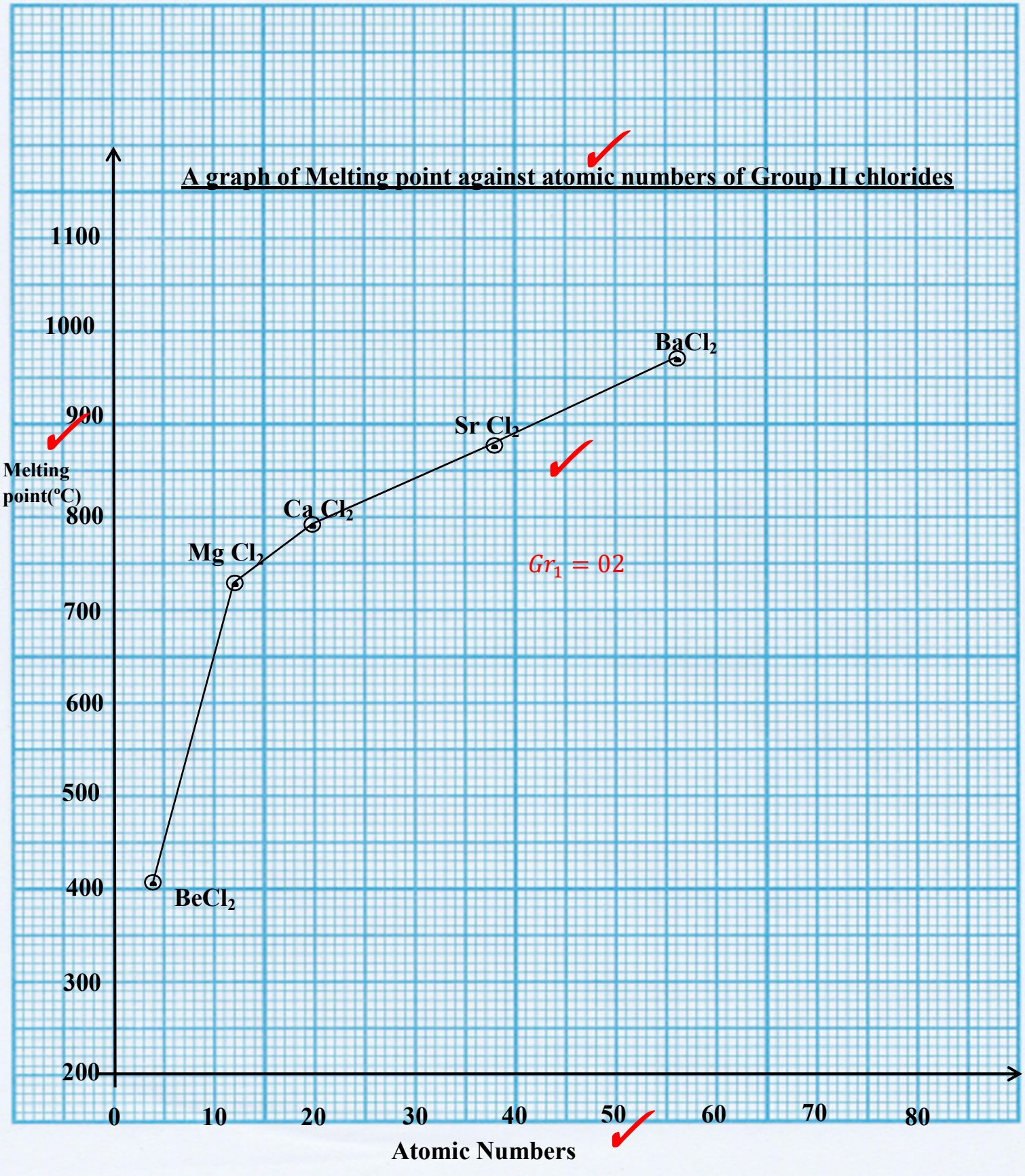
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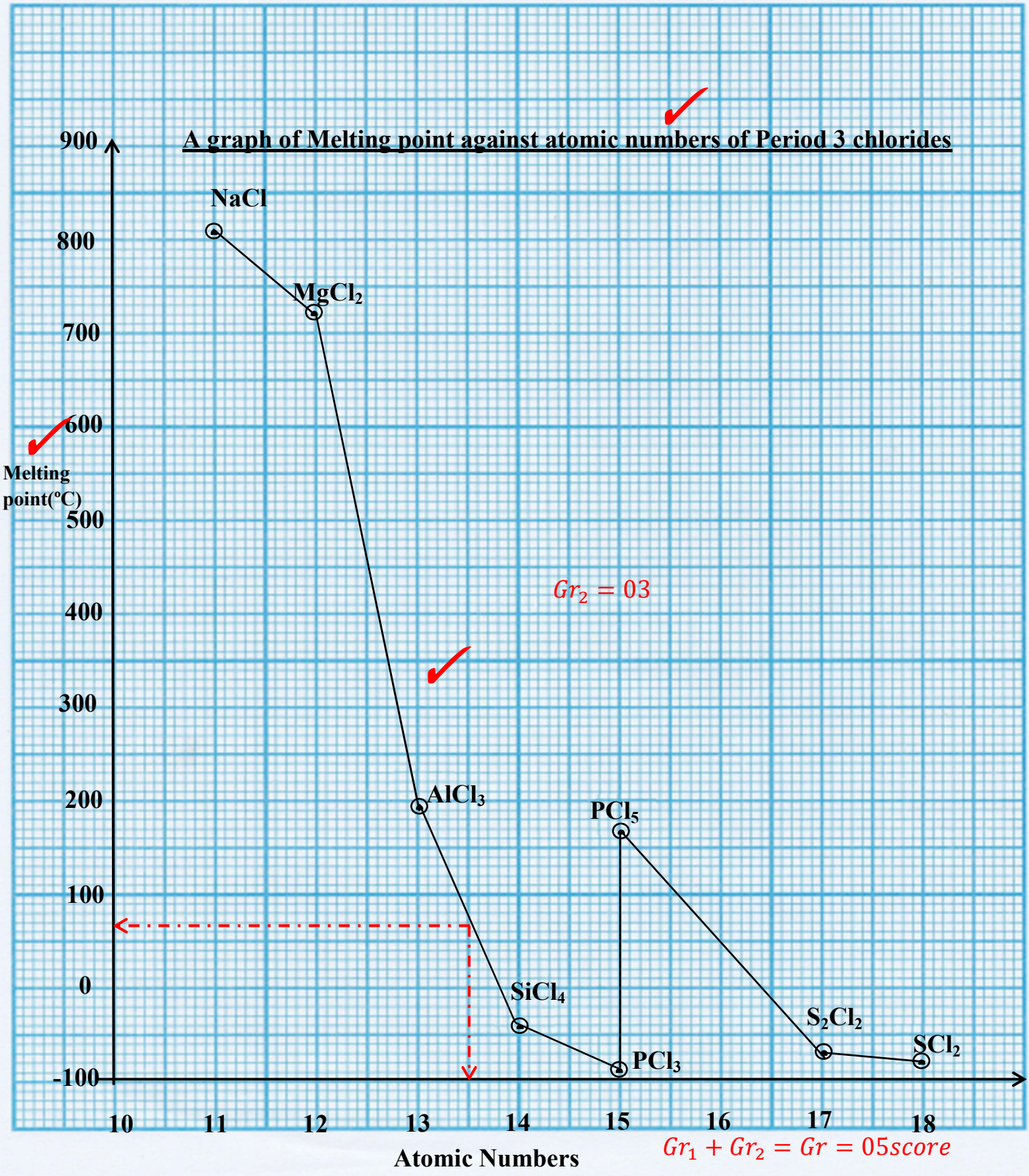
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3. **Trend down group (II) chlorides;**

Melting point ***increases*** *from beryllium chloride to barium chloride* Tr_1

Reason;

The ***cationic radius increases****,* ***charge density*** *and* ***polarising power*** R_1 *of the cations* ***decrease****, and* ***ionic character increases****, requiring an increasing amount of energy to* ***break the increasingly strong ionic bonds.*** R_1

Irregularities; R_1

The melting point of ***beryllium chloride*** *is* ***very low*** *compared to other chlorides because the beryllium ion has a* ***very small ionic radius****,* ***high polarising power*** *and* ***high charge density*** *making beryllium chloride predominantly* ***covalent****. A low amount of energy is required to* ***break the covalent bond****.* R_1

Trend across period 3 chlorides; Tr_2

Melting points of the chlorides generally ***decrease*** *from sodium chloride to sulphur dichloride.*

Reason; R_2

Sodium chloride and magnesium chloride have ***giant ionic structures*** *held by* ***strong ionic bonds*** *which require a* ***high amount of energy to break****. The decrease in melting point from sodium chloride to magnesium chloride is because magnesium ion has a* ***smaller ionic radius****,* ***higher charge density*** *and* ***higher polarising power*** R_2 *than sodium ion making magnesium chloride* ***less ionic*** *than sodium chloride. Aluminium chloride has a* ***lower melting point*** R_2 *than magnesium chloride because among the cations, the aluminium ion has the* ***smallest ionic radius****, a* ***very high charge density*** *and* ***high polarising power****. Aluminium chloride is therefore predominantly* ***covalent****. Silicon tetrachloride is* ***covalent*** *and its molecules are held by* ***weak Van der Waals' forces of attraction that require a low amount of energy to break*** R_2 *hence a low melting point. Phosphorus trichloride has a* ***simple molecular structure*** *thus a low melting point.* R_2 *Disulphur dichloride and sulphur dichloride form simple molecular structures held by weak Van der Waals' forces of attraction.*

Irregularities; R_2

Phosphorus pentachloride has an ***abnormally high*** *melting point because at ordinary temperatures it consists of* PCl_4^+ *+ and* PCl_6^+ *ions hence exhibiting ionic character.*

$Tr_1 + Tr_2 + 4Tr_1 + 4Tr_2 = T_r = 08scores$

Evaluation and choice; E_S

The chlorides $BaCl_2$*,* $SrCl_2$*,* $CaCl_2$*,* $MgCl_2$*, and NaCl are all* ***ionic*** *compounds, due to their* ***very high melting points****, which indicate strong electrostatic forces between metal cations and chloride anions.*

However,

$CaCl_2$ *and* $MgCl_2$ *are ideal for* ***water treatment*** *because they are* ***highly soluble*** *and* ***less toxic****, where they effectively release metal ions that remove impurities through precipitation.* E_d

$CaCl_2$ *is the most suitable chloride in* ***construction purposes*** *for stabilizing concrete because it is* ***hygroscopic****, absorbs moisture from the environment, and accelerates cement setting, thereby increasing the strength and durability of the concrete structure.* E_d

$BaCl_2$ *and* $SrCl_2$*, are the most stable for* ***heat-resistant materials and casting molds*** *used in metal extraction processes. due to* ***highest melting points*** *and* ***strongest ionic bonds*** E_d *among all chlorides listed, they can withstand extremely high temperatures without decomposing,*

making them reliable choices for high-heat industrial applications.

$1E_S + 3E_d = E_v = 04scores$

Prediction of melting point, nature and industrial suitability of chloride X;

*Chloride of element X exists between $AlCl_3$ and its melting point is likely to be between **192** and **−68 °C,** approximately **70°C.*** Pr

*Likely properties of XCl_n; **partial covalent molecular character** rather than a full ionic lattice. Possibly a **low-molecular** covalent chloride.* Pr

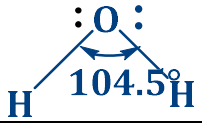
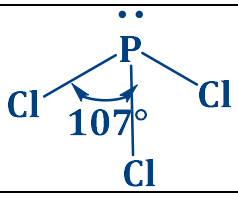
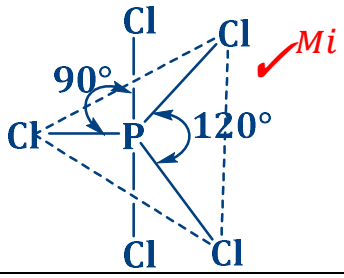
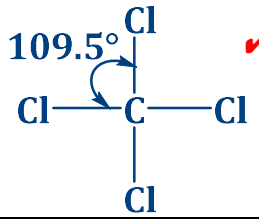
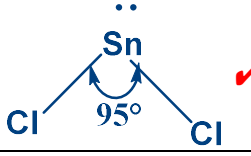
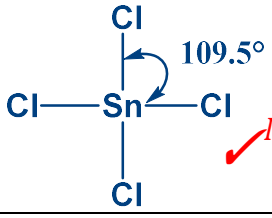
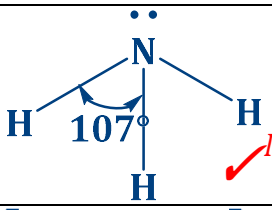
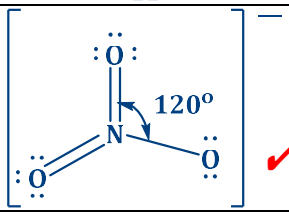
Industrial suitability of XCl_n;

*X is **unsuitable** for water treatment, concrete stabilization and heat-resistant structural materials since it is **covalent.*** Pr

*X can be used as a Lewis acid (like $AlCl_3$) in **organic synthesis** or as an **intermediate** in chemical manufacture, but not as a refractory or concrete stabilizer.* Pr

$3Pr = P_r = 03scores$

4. Molecular shape, bond type, and bond angle;

Name of substance	Molecular structure	Name of molecular shape	Bond type	Bond angle
Water	 ✓*Mi*	*bent shape/ V-shape* ✓*Ni*	*Polar covalent* ✓*Bi*	104.5° ✓*Ai*
Phosphorus trichloride	 ✓*Mi*	*trigonal pyramidal* ✓*Ni*	*Polar covalent* ✓*Bi*	107° ✓*Ai*
Phosphorus pentachloride	 ✓*Mi*	*trigonal bipyramidal* ✓*Ni*	*Covalent* ✓*Bi*	90° and 120°. ✓*Ai*
Carbon tetrachloride	 ✓*Mi*	*tetrahedral* ✓*Ni*	*Non-polar covalent* ✓*Bi*	109.5°. ✓*Ai*
Tin(II) chloride	 ✓*Mi*	Bent shape/ V-shape ✓*Ni*	*Ionic with covalent character* ✓*Bi*	95° ✓*Ai*
Tin(IV) chloride	 ✓*Mi*	*tetrahedral* ✓*Ni*	*Pure covalent* ✓*Bi*	109.5°. ✓*Ai*
Ammonia	 ✓*Mi*	*trigonal pyramidal* ✓*Ni*	*Polar covalent* ✓*Bi*	107° ✓*Ai*
Nitrate ion	 ✓*Mi*	*trigonal planar* ✓*Ni*	*Covalent with resonance delocalization* ✓*Bi*	120° ✓*Ai*

$9Mi + 9Ni + 9Bi + 9Ai = V_p = 08 scorers$

Account for differences in melting point, solubility, and electrical conductivity;

*The **differences in solubility, melting point, and conductivity** among these compounds are due to the type of bonding and molecular structure.*

Ionic compounds *($SnCl_2$, nitrate salts) have **high melting points** and **conduct electricity when molten or in solution**, while **covalent molecular compounds** (CCl_4, PCl_3, $SnCl_4$) have **low melting points**, are **non-conductors**, and are **insoluble** in water.*

Hydrogen bonding *in water and ammonia leads to **strong intermolecular attractions**, giving them higher boiling points than similar covalent molecules like PCl_3 and PCl_5.*

$3R_3 = R_3 = 03scores$

Comparison of boiling points and solubility of water and ammonia to those of phosphorus trichloride, phosphorus pentachloride and carbon tetrachloride;

*Water is a **bent polar molecule** with an angle of **104.5°**, and ammonia is **trigonal pyramidal** with an angle of **107°**; both have **polar covalent bonds** and form **hydrogen bonds** between molecules. Hydrogen bonding gives them **high boiling points** and makes them **highly soluble in water**.*

*In contrast, **PCl_3** (trigonal pyramidal, 107°) and **PCl_5** (trigonal bipyramidal, 90° & 120°) are **covalent** and **cannot form hydrogen bonds**. They have **low melting points**, are **liquid or volatile solids**, and **do not dissolve** in water because they hydrolyse, producing **HCl** and **phosphoric acids**.*

$3R_4 = R_4 = 03scores$

Comparison of electrical conductivity and melting points of tin(II) and tin(IV) chloride;

*Tin(II) chloride ($SnCl_2$) is **ionic with some covalent character**, forming a **bent structure** (≈95°) and a **solid** at room temperature. It **conducts electricity when molten** because it contains **mobile ions (Sn^{2+} and Cl^-)**.*

*Tin(IV) chloride ($SnCl_4$), on the other hand, is **tetrahedral (109.5°)** and **purely covalent**, existing as a **molecular liquid** that **does not conduct electricity**. Its weak molecular forces give it a **lower melting point** than $SnCl_2$.*

$3R_5 = R_5 = 03scores$

$R_3 + R_4 + R_5 = R = 09scores$

Evaluation of the environmental impacts of using nitrate ions in fertilizers;

*In fertilizer use, **nitrate ions (NO_3^-)** improve plant growth.*

*However when overused, cause **environmental pollution** since Nitrates are **highly soluble** and can **leach into groundwater**, causing **eutrophication** and **oxygen depletion** in water bodies, leading to fish deaths and **blue-baby syndrome** in humans. Mitigated by controlled, fertilizer application, and use of organic manure.*

$1di + 1de + 1dm = D_3 = 03\ Scores$

5. **Formulae of W ;**

$C_xH_y(g) + \left(x + \frac{y}{4}\right) O_2(g) \longrightarrow xCO_2(g) + \frac{y}{2}H_2O(g)$

$Volume\ of\ hydrocarbon = 20cm^3$

$Volume\ of\ carbon\ diode\ produced = 110 - 30 = \mathbf{80cm^3}$

$Volume\ of\ oxygen\ used\ up\ in\ reaction = original\ volume - volume\ unreacted$

$= 150 - 30 = \mathbf{120cm^3}$ ✓CA

$Volume\ of\ hydrocarbon \times x = volume\ of\ carbondioxide$

$\Longrightarrow 20x = 80$

$\therefore x = {}^{80}/_{20} = \mathbf{4}$

$Also\ volume\ of\ hydrocarbon \times \left(x + \frac{y}{4}\right) = volume\ of\ oxygen\ used\ up$ ✓CA

$\Longrightarrow 20\left(4 + \frac{y}{4}\right) = 120$

$\therefore\ y = (6 - 4) \times 4 = 8$

$\therefore$ ***The formula of W is*** $\mathbf{C_4H_8}$ ✓CA

Possible isomers, identity of W with reason(s)

$\mathbf{CH_3CH_2CH = CH_2}$

$\mathbf{CH_3CH = CHCH_3}$ ✓CA

Since ozonolysis of W produced only one compound, it indicates that it has a symmetrical carbon-carbon double bond.

$\mathbf{CH_3CH = CHCH_3} \xrightarrow[2.\ H_2O, Zn, CH_3COOH]{1.\ O_3, CCl_4, < 20^0C} \mathbf{2CH_3CHO}$ ***, One compound produced is ethanal***

Therefore, W is $\mathbf{CH3CH = CHCH3}$ ***but-2-ene*** ✓CA

Comfirmatory Test for Carbon dioxide; ✓I

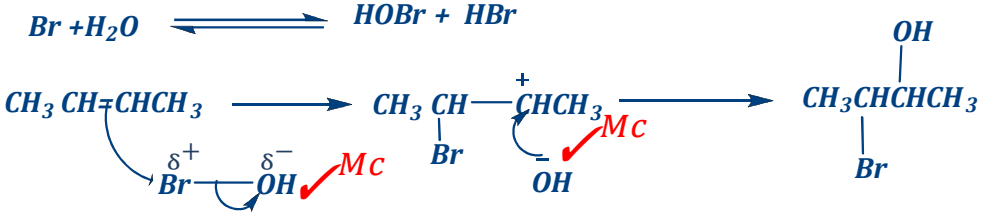
If the gas is CO_2, KOH will absorb it so the ***limewater remains clear*** *(no milky cloud), because CO_2 was removed by the KOH beforehand. But the gas is not absorbed by KOH, the limewater will still* ***go milky****.* ✓I

$5CA + 2I = \text{F} = 06\ Scores$

Mechanism for reaction between W and bromine water;

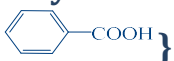
$\mathbf{CH_3CH = CHCH_3} \xrightarrow{Br_2/H_2O} \mathbf{CH_3CHBrCH(OH)CH_3 + HBr}$

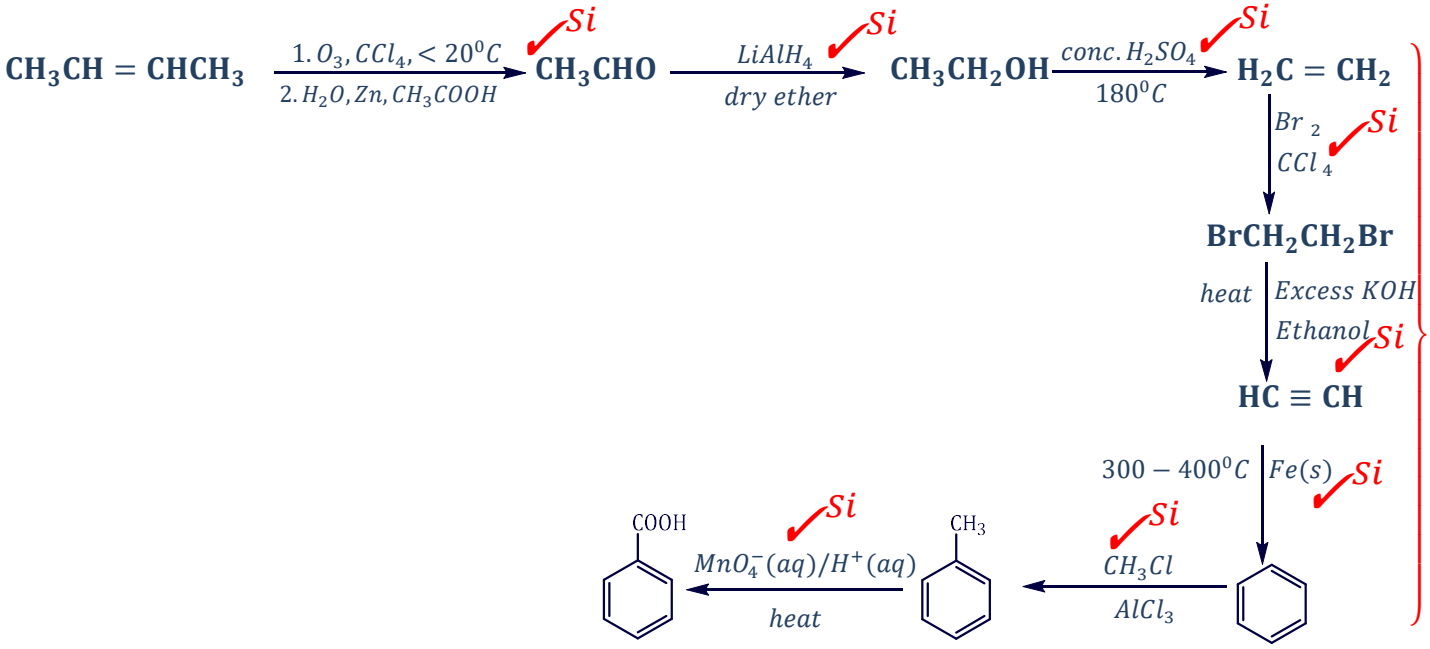
Mechanism



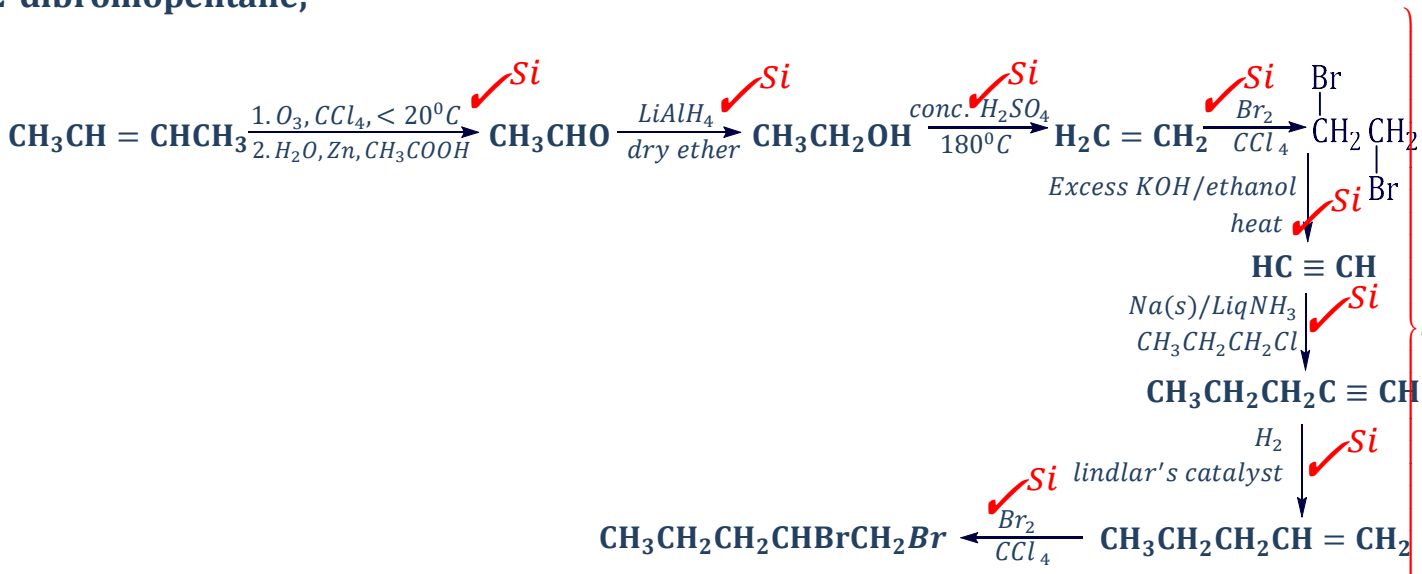
$2Mc = \text{Mc} = 02\ Scores$

Synthetic **pathways to related products;**

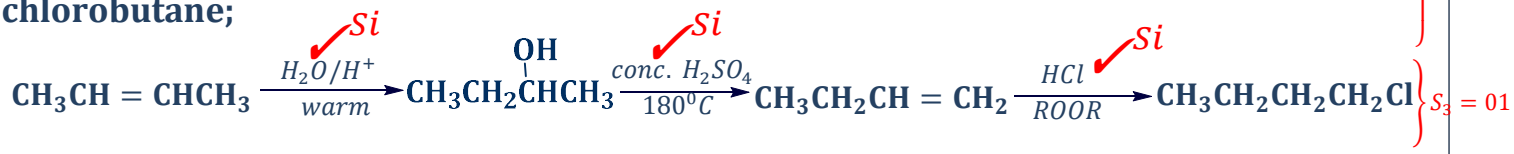
Benzoic acid,{  **};**



1,2-dibromopentane;



1-chlorobutane;



$S_1 + S_2 + S_3 = S = 11\ Scores$

Environmental impacts of producing above compounds in industry;

When burnt, they give off poisonous gases such as carbon monoxide (CO), carbon dioxide (CO_2), hydrogen chloride (HCl), and hydrogen bromide, hence lead to ***air pollution***, ***acid rain***, ***respiratory problems***, *and* ***global warming***. *The acid gases can also corrode metals and destroy plant leaves. Mitigated avoid* ***open-air burning*** *and instead use* ***safe chemical disposal methods***.

di de dm

$1di + 1de + 1dm = D_3 = 03\ Scores$

6. **Formulae of P ;**

$\boldsymbol{Mass\ of\ carbondiode\ formed = 8.8g}$ $\boldsymbol{and\ mass\ of\ water\ formed = 1.8}$

$\boldsymbol{mass\ of\ carbon = \frac{12}{44} \times 8.8 = 2.4g}$

$\boldsymbol{mass\ hydrogen = \frac{2}{18} \times 1.8 = \ 0.2g}$

Element	C	H
%composition	*2.4*	*0.2*
Moles	$\frac{2.4}{12} = \mathbf{0.2}$	$\frac{0.2}{1} = \mathbf{0.2}$
Simplest ratio	$\frac{0.2}{0.2} = \mathbf{1}$	$\frac{0.2}{0.2} = \mathbf{1}$

CA

$\therefore$ *Empirical formula is* **CH**

Mass of p vaporised $= 0.100g$

At a Temperature $T = 273 + 273 = 546K$

At a Pressure $P = \left(\frac{734}{760} \times 1\right)\ atm$

At a Volume $V = 4.46 \times 10^{-2} dm^3$

Note: 1. *we use;* $\boldsymbol{R = 0.0821}$ L · atm/(mol · K), *when, P is in atm and V is in litre/*dm^3

2. *we use;* $\boldsymbol{R = 8.314\ J/(molK)}$, *when, P is in Pa and Volume is in* m^3

From ideal gas law $PV = nRT$

$$PV = \frac{m}{M_r}RT$$

$$\Longrightarrow M_r = \frac{mRT}{PV} = \frac{0.100 \times 0.0821 \times 546}{\left(\frac{734}{760} \times 1\right) \times 4.46 \times 10^{-2}} = \frac{4.48266}{0.04307} \approx \mathbf{104g}$$

CA

$\therefore$ $\mathbf{(CH)_n = 104}$

$(12 \times 1 \times n) + (1 \times 1 \times n) = 104$

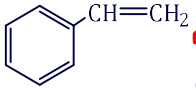
$13n = 104$

$n = \frac{104}{13} = 8$

$\therefore$ *Molecular formula is* $\mathbf{C_8H_8}$

CA

Structure and IUPAC Name;



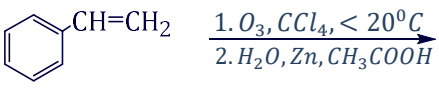
CA

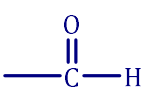
ethenylbenzene CA

Structural isomers;

P *has no other strural isomers.* I

Identity of P and mixture Q;

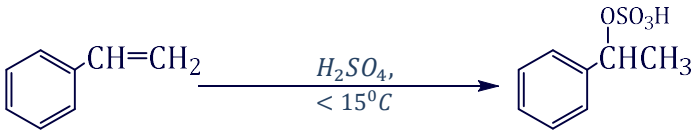


Q *is a mixture of benzylaldehyde and methanal with functional group*  I

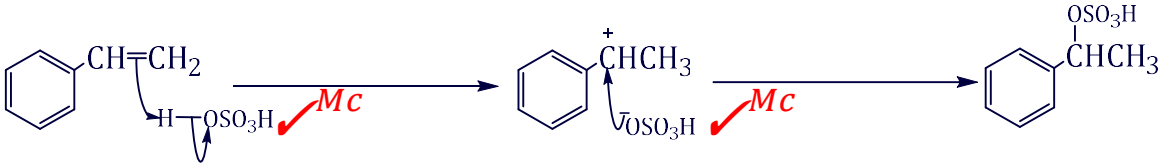
$5CA + 2I = F = 06\ Scores$

Mechanisms for reactions of P with bromine and concentrated sulphuric acid;

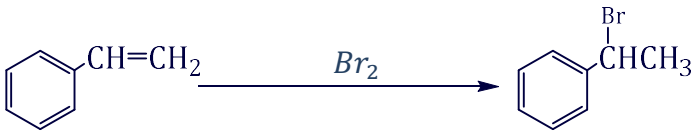
With concentrated sulphuric acid;

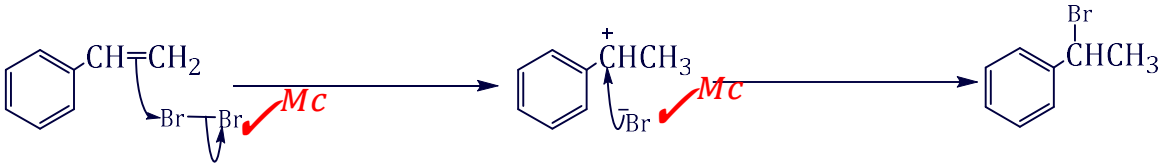


Mechanisim



With bromine;



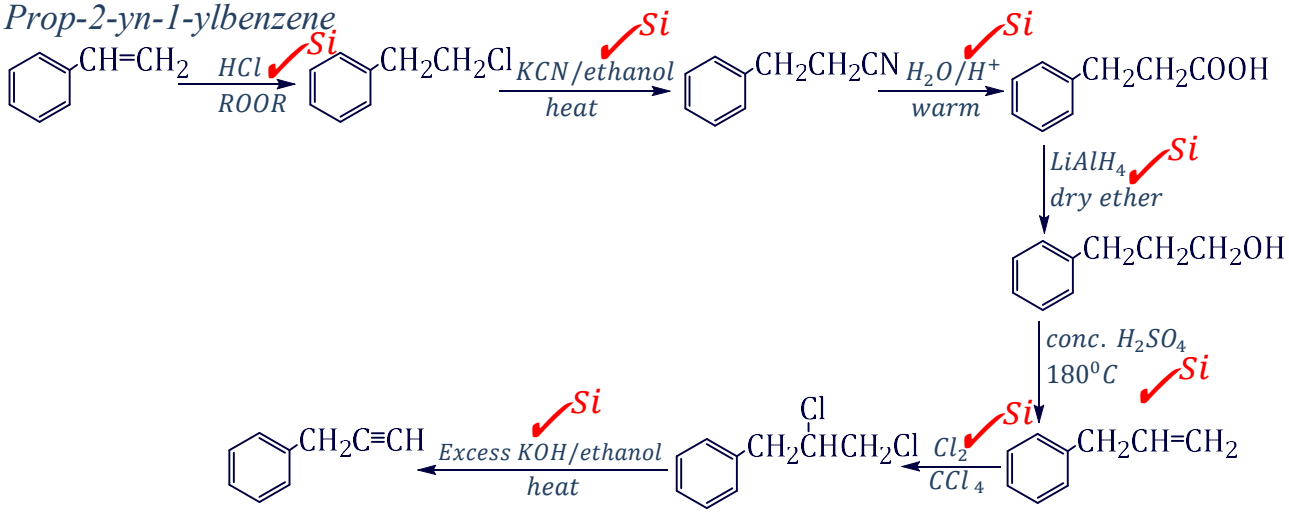


$M_C = 02$

$4Mc = Mc = 02\ Scores$

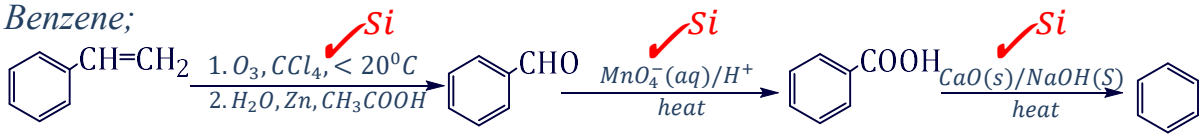
Synthetic pathways from Albertine oil fractions P to synthesis industrial compounds ;

Prop-2-yn-1-ylbenzene



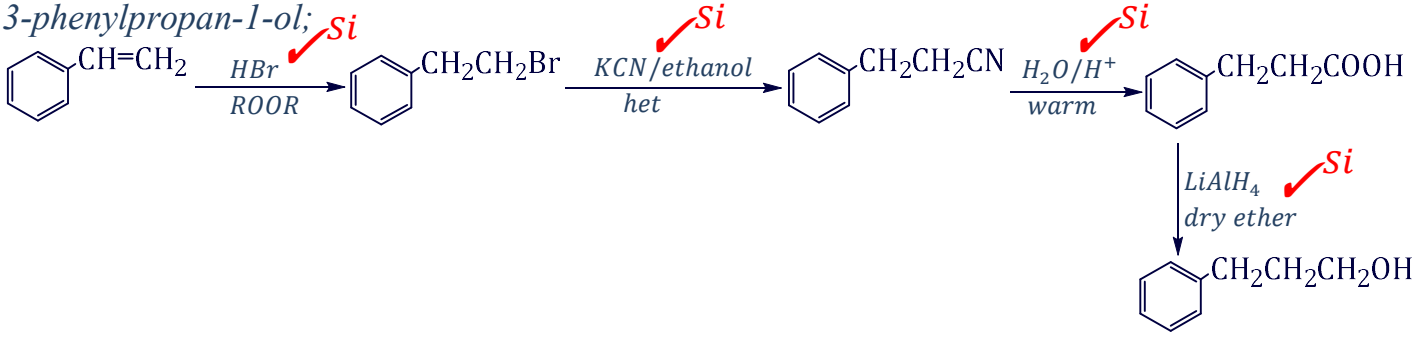
$S_1 = 05$

Benzene;



$S_3 = 01$

3-phenylpropan-1-ol;



Si Si Si Si

$S_1 = 05$

$S_1 + S_2 + S_3 = S = 03\ Scores$

Environmental impacts of producing above compounds in industry;

*When burnt, they give off poisonous gases such as carbon monoxide (CO), carbon dioxide (CO_2), hydrogen chloride (HCl), and hydrogen bromide, hence lead to **air pollution**, **acid rain**, **respiratory problems**, and **global warming**. The acid gases can also corrode metals and destroy plant leaves. Mitigated avoid **open-air burning** and instead use **safe chemical disposal methods**.*

di de dm

1di+1de+1dm=D₃=03 Scores

END

2025

It is All about mindset change